

Jonathan Zhao

jonnny80@gmail.com | 929-428-1109 | saloushe.github.io | github.com/saloushe | linkedin.com/in/joezhao888

EDUCATION

Massachusetts Institute of Technology (MIT)

June 2026

Bachelor's Degree in Computer Science and Engineering

Cambridge, MA

- Relevant Coursework: Computer Systems Engineering, Operating Systems Engineering, Database Systems, Design and Analysis of Algorithms, Programming in C and Assembly, Software Design, Natural Language Processing, AI, Decision Making, and Society

WORK EXPERIENCE

Simons Electron Microscopy Center

May 2025 – August 2025

Software Engineering Intern

New York, NY

- Designed Cryo-Insight, a Python library enabling Leginon users to convert cryo-EM datasets into EMinisight schema format for downstream analysis workflows
- Extended EMinisight with additional data aggregation, reporting, and visualization features

Duet

Nov 2024 – Feb 2025

Backend Engineer Intern

Remote

- Built a MusicXML parsing and analysis pipeline in Python (from-scratch and Music21-based implementations) extracting structural score data (parts, measures, barlines, repeats, time signatures) and segmentation utilities for an AI-based sheet music writing copilot

MIT Media Lab

Oct 2022 – June 2023

Undergraduate Researcher, MIT Game Lab

Cambridge, MA

- Designed player-facing ARG puzzles teaching DEI principles to high school students, blending physical puzzle mechanics with virtual presentation

Feil Family Brain & Mind Research Institute

April 2019 – November 2021

Researcher, Gang Wang Lab

New York, NY

- Researched voltage-gated L-type calcium channels in neuropsychiatric disorders in early-stage Alzheimer's Disease; co-discovered role of L-type channels and amygdala NPY neurons in anxiety-related behaviors

PROJECTS

GoDB — Relational Database Engine in Go

Spring 2026

- Built a relational database engine in Go across 4 labs: heap file storage, a concurrent CAS-based buffer pool, and a Volcano-style execution engine with 4 join algorithms (hash, sort-merge, block nested-loop, index nested-loop), grouped aggregation, and full CRUD executors
- Implemented Strong Strict 2PL transactions with multi-granularity (table/tuple) locking and wait-for-graph deadlock detection, plus a rule-based query optimizer (predicate pushdown, priority-ranked physical plan selection)
- Built ARIES-style crash recovery: a checksummed write-ahead log with torn-write detection, a double-buffered log manager enabling group commit, and Analysis/Redo/Undo recovery with fuzzy checkpointing

WaitLess — Real-Time Nightlife Crowdsourcing Platform

Fall 2025

- Built full-stack venue crowdsourcing platform (React, Node.js/Express, MongoDB) with Socket.io real-time updates, configurable alert subscriptions triggered by user-defined wait time and crowd conditions
- Implemented geospatial indexing for location-aware queries and background snapshots aggregating crowd stats for peak time forecasting

Why Transformers Miss Toxicity — NLP Interpretability Research

Spring 2025

- Fine-tuned BERT on 322K+ examples across 3 toxicity corpora (90.5% accuracy, 0.76 F1); trained a 128-unit autoencoder over final-layer CLS embeddings to surface toxicity-correlated features (Pearson $r = 0.87$), then causally tested them through feature and attention-head ablation — finding the representation redundant rather than localized

OTHER EXPERIENCE

Sigma Alpha Epsilon Fraternity, Mass Iota-Tau Chapter

Jan 2025 – Jan 2026

President

Cambridge, MA

- Managed and led a 30+ member fraternity chapter; oversaw a \$100,000 annual budget for event planning and chapter operations

SKILLS & INTERESTS

- Technical: Go, Python (pandas, NumPy, SQLAlchemy), MERN stack (JS/TS, Node.js, MongoDB, Express), React, Socket.io, ML Engineering (PyTorch, scikit-learn, HuggingFace), C, Assembly, AWS, SQL, Agile
- Languages: English (Fluent), Mandarin Chinese (Fluent), French (Basic)
- Interests: Game Design, Modding Video Games, Poker, Boxing, Film, Philosophy, Motorsports